UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 003 · Particular integral Type 1 ($X = e^{ax}$) · compiled 15 October 2026

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1 Short notes

1 · The inverse differential operator and the particular integral
For $f(D)y = X$ ($X \neq 0$) the general solution is $y = \mathrm{CF} + \mathrm{PI}$. The inverse operator $\frac{1}{f(D)}X$ is the function of x, free from arbitrary constants, that gives X when $f(D)$ acts on it: $f(D)\{\frac{1}{f(D)}X\} = X$. Theorem: putting $y = \frac{1}{f(D)}X$ in the equation gives $X = X$, so $\frac{1}{f(D)}X$ is a particular solution, the PI.

2 · Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$
$\frac{1}{D}X = \int X\,dx$ and $\frac{1}{D^2}X = \iint X\,dx\,dx$ (no constants). $\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx$, $\frac{1}{D+a}X = e^{-ax}\int e^{ax}X\,dx$. For $(D-a)(D-b)$ apply the factors one at a time or use partial fractions. Write the operator before the function.

3 · Type 1: $X = e^{ax}$ and the cases of failure
$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}$ if $f(a) \neq 0$. If $f(a) = 0$: $\frac{xe^{ax}}{f'(a)}$; if $f'(a) = 0$ too: $\frac{x^2e^{ax}}{f''(a)}$; then $\frac{x^3e^{ax}}{f'''(a)}$, and so on. The power of x is the multiplicity of a as an AE root. Examples: $\frac{e^{-2x}}{D^2+4D+4} = \frac{x^2e^{-2x}}{2}$; $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ (ESA ×3).

4 · Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x
Constants: $\frac{k}{f(D)} = \frac{k}{f(0)}$ (if $f(0) = 0$: $\frac{xk}{f'(0)}$). $\frac{\sinh ax}{f(D)} = \frac{1}{2}[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}]$, $\cosh$ with $+$. $\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}$, $\frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)}$. CW P11: $\frac{x^2e^x}{6} - \frac{e^{-x}}{4}$ (the sheet's $\frac{x^2e^x}{5}$ is wrong); CW P12: $\frac{3}{8}e^{-2x} - \frac{2}{9}x^3e^{-x} - \frac{9}{4}$.

2 Keyword index

19 terms; the number in brackets is the topic.

$\cosh ax$ (4). $\frac{e^{ax}+e^{-ax}}{2}$.

$\frac{1}{D}$ (2). Integration without a constant.

$\frac{1}{D-a}$ (2). $e^{ax}\int e^{-ax}X\,dx$.

$\sinh ax$ (4). $\frac{e^{ax}-e^{-ax}}{2}$.

a^x (4). $e^{x\log a}$: replace D by $\log a$.

e^{0x} (4). A constant: replace D by 0.

$f'(a)$ (3). Derivative of $f(D)$ w.r.t. D, at $D=a$.

Case of failure (3). $f(a)=0$; a is an AE root.

Complementary function (1). General solution of $f(D)y=0$.

Failure ladder (3). $\frac{e^{ax}}{f(a)}\to\frac{xe^{ax}}{f'(a)}\to\frac{x^2e^{ax}}{f''(a)}\to\cdots$

General solution (1). $y=y_c+y_p$.

Inverse differential operator (1). $\frac{1}{f(D)}$; undoes $f(D)$.

Multiplicity (3). Power of x in a failed PI.

Non-homogeneous LDE (1). $f(D)y=X$ with $X\neq 0$.

Partial fractions (2). Split $\frac{1}{(D-a)(D-b)}$ into simple factors.

Particular integral (1). $\frac{1}{f(D)}X$; no arbitrary constants.

PI$_1$, PI$_2$ (4). PIs of the separate terms of X; add them.

Resonance-like growth (3). Extra factor x^k when X matches a CF term.

Type 1 (3). $X=e^{ax}$: replace D by a.

3 Rapid recall

1	General solution of $f(D)y = X$?	$y = \text{CF} + \text{PI} = y_c + y_p$.
2	PI in operator form?	$\frac{1}{f(D)}X$.
3	Does the PI contain arbitrary constants?	No; they are all in the CF.
4	$\frac{1}{D}X$?	$\int X\,dx$, no constant.
5	$\frac{1}{D-a}X$?	$e^{ax}\int e^{-ax}X\,dx$.
6	$\frac{1}{f(D)}e^{ax}$, $f(a)\neq 0$?	$\frac{e^{ax}}{f(a)}$.
7	$\frac{1}{f(D)}e^{ax}$, $f(a)=0$, $f'(a)\neq 0$?	$x\frac{e^{ax}}{f'(a)}$.
8	When is $f(a)=0$?	When $D-a$ is a factor of $f(D)$ (a an AE root).
9	$\frac{e^{-2x}}{D^2+4D+4}$?	$\frac{x^2e^{-2x}}{2}$.
10	$\frac{1}{D^2+2D+5}$ (RHS 1)?	$\frac{1}{5}$.
11	$\frac{8e^{2x}}{(D-2)^2}$?	$4x^2e^{2x}$.
12	$\frac{e^{2x}}{D^3-6D^2+11D-6}$?	$-xe^{2x}$.
13	$\frac{a^x}{f(D)}$?	$\frac{a^x}{f(\log a)}$.
14	$\frac{2^{-x}}{D^2+4}$?	$\frac{2^{-x}}{(\log 2)^2+4}$.
15	$\frac{\cosh ax}{f(D)}$?	$\frac{1}{2}\left[\frac{e^{ax}}{f(a)}+\frac{e^{-ax}}{f(-a)}\right]$.
16	PI of $y'''-3y'+2y=2\sinh x$?	$\frac{x^2e^x}{6}-\frac{e^{-x}}{4}$ (CW's $\frac{x^2e^x}{5}$ is wrong).
17	PI of $y''-3y'-4y=e^x+4e^{4x}$?	$-\frac{e^x}{6}+\frac{4}{5}xe^{4x}$.
18	$\frac{1/2}{D^3-2D^2+D}$?	$\frac{x}{2}$ ($f(0)=0$, $f'(0)=1$).

Mnemonic (study aid). *"Fail once, add an x; differentiate the bottom."* The failure ladder in one line (study aid).

Mnemonic (study aid). *"a^x hides $\log a$."* Replace D by the exponent of e, not by the base (study aid).

4 Last 10 minutes before the exam

Top facts

1. $y = y_c + y_p$; PI $= \frac{1}{f(D)}X$, free of arbitrary constants.
2. $\frac{1}{D}X = \int X\,dx$; $\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx$.
3. $\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}$, $f(a) \neq 0$; else $\frac{xe^{ax}}{f'(a)}$; else $\frac{x^2e^{ax}}{f''(a)}$; ...
4. Failure happens exactly when a is a root of the AE: check against the CF first.
5. Constants: $D \to 0$. $\sinh$ / $\cosh$: $D \to \pm a$. $a^{\pm x}$: $D \to \pm \log a$.
6. ESA: $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ (three papers); $\frac{2^{-x}}{D^2+4} = \frac{2^{-x}}{(\log 2)^2+4}$.
7. CW P11: $\frac{x^2e^x}{6}$, not the sheet's $\frac{x^2e^x}{5}$.

Top formulas

E1.2 · General solution of $f(D)y = X$ ESA ★★★

$$y = y_c + y_p = \text{CF} + \frac{1}{f(D)}X$$

GIVES The complete solution.

APPLIES $X \neq 0$; the CF from the auxiliary equation (class 002).

NOTE CF has n arbitrary constants; the PI has none.

S003·6 · S003·12 · T003 00:05:32 · T003 00:06:07

E3.1 · Type 1 rule ESA ★★★

$$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}, \qquad f(a) \neq 0$$

GIVES The PI of an exponential.

APPLIES $X = e^{ax}$, a positive, negative or zero.

NOTE Replace D by a. For e^{-ax} replace D by $-a$: $\frac{e^{-ax}}{f(-a)}$ (T003 00:10:51).

S003·8 · S003·13 · T003 00:10:15 · T003 00:10:51 · T004a 00:00:41

E3.2 · First case of failure ESA ★★★

$$f(a) = 0: \quad \frac{1}{f(D)}e^{ax} = x\,\frac{e^{ax}}{f'(a)}, \qquad f'(a) \neq 0$$

GIVES The PI when a is a simple root of $f(m) = 0$.

APPLIES $D - a$ is a factor of $f(D)$ once.

NOTE $f'(D) = \frac{d}{dD}f(D)$; x is written outside the operator (T003 00:11:51).

S003·8 · S003·13 · T003 00:11:35 · T003 00:12:08 · T004a 00:01:20 · ESA Jan-24 s2 2d

E3.3 · Second case of failure

ESA ★★★

$$f(a) = f'(a) = 0: \quad \frac{1}{f(D)} e^{ax} = x^2 \frac{e^{ax}}{f''(a)}, \qquad f''(a) \neq 0$$

GIVES The PI when a is a double root.

APPLIES $(D-a)^2$ divides $f(D)$.

NOTE ESA Dec-23 3a: $\frac{8e^{2x}}{(D-2)^2} = \frac{8x^2e^{2x}}{2} = 4x^2e^{2x}$.

S003·8 · S003·13 · T003 00:12:53 · T004a 00:01:41 · ESA Dec-23 3a

E4.1 · Constant right-hand side

ESA ★★

$$\frac{k}{f(D)} = k\,\frac{e^{0x}}{f(D)} = \frac{k}{f(0)}, \qquad f(0) \neq 0$$

GIVES The PI of a constant.

APPLIES $X = k$.

NOTE If $f(0) = 0$: $\frac{xk}{f'(0)}$, then $\frac{x^2k}{f''(0)}$, ... (T003 00:14:25). Slide 003·14: $\frac{1}{D^2+2D+5} = \frac{1}{5}$.

S003·9 · S003·14 · T003 00:13:50 · T003 00:14:25 · T004a 00:02:06 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E4.2 · $\sinh ax$

ESA ★★

$$\frac{\sinh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$$

GIVES The PI of $\sinh ax$.

APPLIES $f(a) \neq 0$, $f(-a) \neq 0$; otherwise use the failure ladder on that term.

NOTE $\sinh ax = \frac{e^{ax}-e^{-ax}}{2}$. Slide 003·16: $2\sinh x = e^x - e^{-x}$.

S003·9 · T003 00:14:55 · T003 00:15:19 · CW U2 P11

E4.4 · a^x and a^{-x}

ESA ★★

$$\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}, \qquad \frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)}$$

GIVES The PI of a power of a constant.

APPLIES $a > 0$, $f(\pm \log a) \neq 0$.

NOTE $a^x = e^{x \log a}$: the exponent is $\log a$ (natural log). ESA Jan-24 s1 2d and s4 2c ask 2^{-x}.

S003·10 · T003 00:15:54 · T003 00:16:34 · T003 00:16:48 · ESA Jan-24 s1 2d · ESA Jan-24 s4 2c